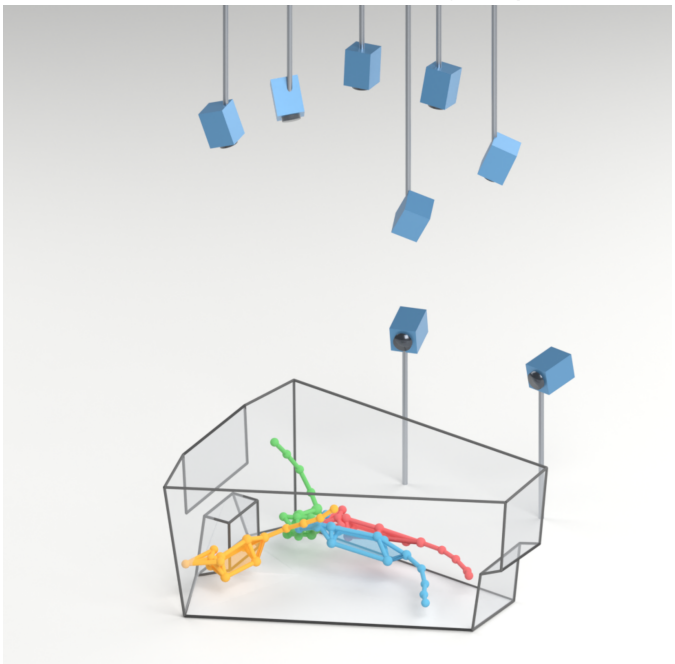
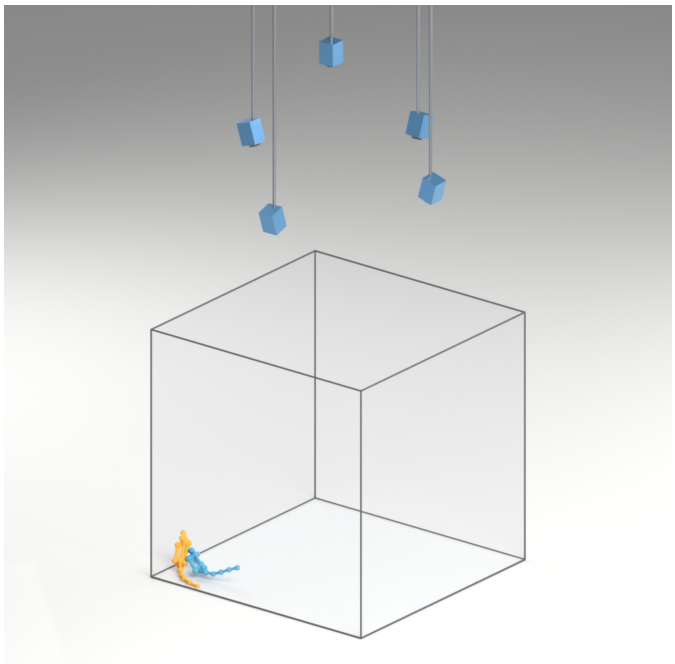
a Dataset arena visualizations with camera rigs and poses

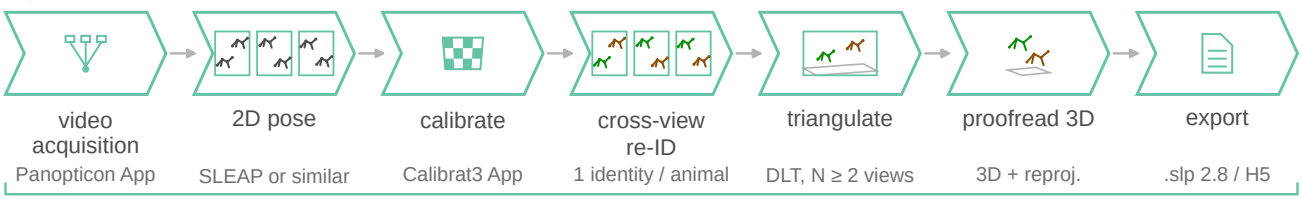


SLAP-2M



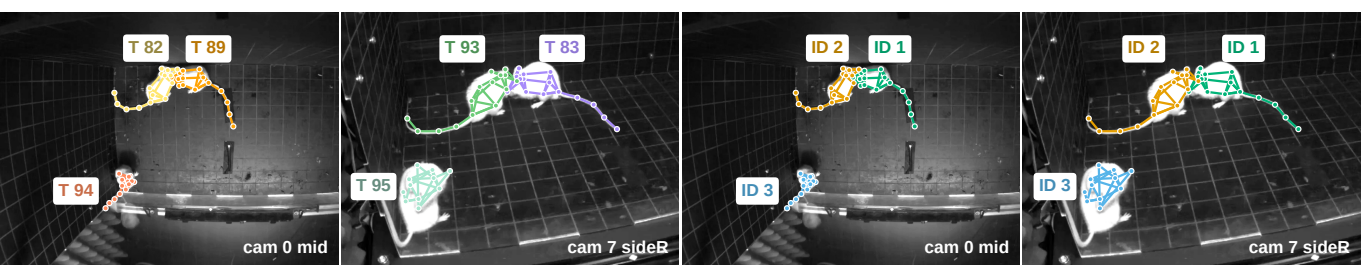
Mouse-Dyad-10M

b Pipeline

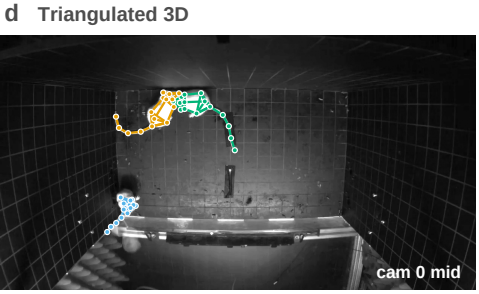


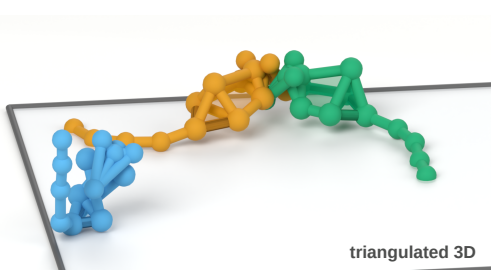
c Cross-view re-identification

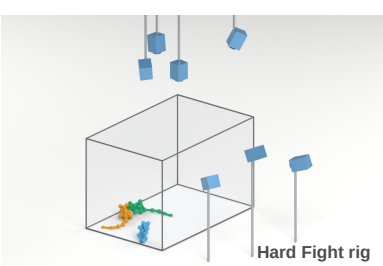
per-camera tracks → LUC3D identities 2 of 8 views



d Triangulated 3D







e Capability comparison

	No installation	Browser-based	Multi-animal	Multi-camera 3D	Cross-view ID	3D proofreading
LUC3D (this work)	✔	✔	✔	✔	✔	✔
SLEAP	–	–	✔	sleap-anipose	–	2D only
Label3D	–	–	–	✔	–	✔
JARVIS (HybridNet)	–	–	–	✔	–	reproj. error
Lightning Pose	–	✔	–	multi-view losses	–	–
DeepLabCut	–	–	✔	pairwise	–	–
Anipose	–	–	–	✔	–	–
DANNCE / SDANNCE	–	–	✔	✔	–	–